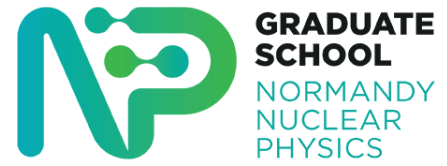




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Tumour Geometric Complexity: A Proof of Concept Towards Explainable Quality Assurance in Radiotherapy Trials

by

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All analyses were conducted on pseudonymised clinical trial data provided under the appropriate trial-level authorisations and data-transfer agreements. Only trial pseudonyms were available to the analysis team, and no directly identifying information was accessed. Data handling and processing were performed within a secure environment and in compliance with applicable ethical, confidentiality, and regulatory requirements.

Generative artificial intelligence was used in the preparation of this thesis to edit the language of the manuscript, covering grammar, phrasing, and the consistency of terminology across chapters; to check the text against the tables and the figures for internal consistency; and to assist with LaTeX formatting. It is declared here in accordance with the Règlement commun des études of the Université de Caen Normandie (2022–2027, §6.2). The tool was Claude (Anthropic), used through the web interface between June and August 2026.

The tool was not used to generate the scientific content of this work. The formulation of the Geometric Volume Coefficient, the computational pipeline, the treatment plans generated in matRad, the numerical results, and their interpretation are my own. No bibliographic reference was taken from the tool; every reference cited was located, read, and verified in the original publication. I take full responsibility for the content of the manuscript, including for any error introduced during editing.

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Abstract

Radiotherapy is an applied physics problem: ionising radiation deposited in heterogeneous anatomy, under geometric and biological constraints. In clinical trials, each treatment plan is checked by a quality-assurance (QA) review before delivery. That review returns a binary verdict. A plan passes or it fails, and when it fails the review cannot say whether the planning was poor or whether the patient’s anatomy put the dosimetric goals out of reach. The two situations call for opposite responses, and current QA does not separate them. It is also manual, expert-dependent, and hard to scale across centres.

This thesis quantifies how difficult a case is from its geometry alone, as the first step towards a QA process that can explain *why* a plan fails. The measure is the Tumour Geometric Complexity, computed through the Geometric Volume Coefficient at the organ and at the patient level. A distance-to-target histogram is partitioned into overlap, near-compression, and separation zones, and the coefficient is built from the delineated contours alone. Its overlap term is purely anatomical; its near-compression term additionally carries the limit of machine deliverability, through the achievable dose gradient. To test the measure against dose, a multi-modality planning study was carried out in matRad for photon, proton, helium-ion, and carbon-ion plans, with particle plans evaluated as RBE-weighted dose. The per-organ delivered doses give an independent check on the geometric measure.

The descriptor was computed for three head-region patients, one of them carried through the full pipeline in detail. Patient-level values ranged from 0.04 for a well-separated case to 1.15 for one in which the optic chiasm was almost fully enclosed by the target. The central result is the agreement between geometry and dose. Ordering the organs at risk by their contour-only coefficient reproduces the ordering of their delivered mean doses, with a Spearman rank correlation between 0.77 and 0.96 across all twelve patient–modality combinations (all $p < 0.01$), and the agreement holds for every modality and every patient. A purely geometric descriptor therefore recovers the dose burden imposed by the anatomy, across the delivery techniques tested. This is the measure a complexity-aware quality assurance can be built on: it lets the binary verdict be opened into four classes, Pass, Fail, Sub-optimal (low complexity), and complexity-limited (high complexity), so that a plan limited by poor planning is distinguished from one limited by the anatomy. That scheme is proposed here as the natural application of the finding.

These findings are a proof of concept established on three cases. The descriptor’s parameters and the four-class scheme both await validation on the full cohort.

Keywords: Radiotherapy, Quality Assurance, Tumour Geometric Complexity, Treatment Planning, matRad, Particle Therapy

Contents

List of Abbreviations	1
1 Introduction	2
1.1 Historical Development of Radiotherapy	2
1.2 Principles of Radiotherapy	2
1.2.1 Radiotherapy Techniques	3
1.3 Clinical Trials and the Role of Quality Assurance	3
1.4 The Current QA Process and Its Limitations	4
1.5 Areas of Improvement	5
1.6 Treatment Planning and Anatomical Complexity	5
1.7 Objective of the Thesis	6
2 Methods	7
2.1 Tumour Geometric Complexity: Formulation and Computation	7
2.1.1 Organ-level Geometric Volume Coefficient	8
2.1.2 Interpretation of the organ-level coefficient	9
2.1.3 Patient-level Coefficient	10
2.1.4 Computation	11
2.2 Multi-Modality Treatment Planning with matRad	12
2.2.1 Patient data and structures	12
2.2.2 The planning workflow	12
2.2.3 Plan configurations and modalities	12
2.3 Agreement Between Geometric Volume Coefficient and Dosimetric Outcomes . .	14
2.4 Proposed Integration of Tumour Geometric Complexity into the QA Framework .	14
3 Results	16
3.1 Tumour Geometric Complexity: Organ-Level and Patient-Level Computation . .	16
3.1.1 Organ-level Geometric Volume Coefficient	16
3.1.2 Distance-to-target histograms	17
3.1.3 Patient-level Coefficient	17
3.1.4 Organ-level Coefficient Across Three Patients	18
3.2 Treatment Planning Results: matRad Multi-Modality Study	19
3.2.1 Dose Distributions	19
3.2.2 Dose–Volume Histograms	20
3.2.3 Mean and Maximum Doses	21

3.3	Agreement Between Geometric Volume Coefficient and Delivered Dose	22
3.3.1	Sensitivity to the Zero-Complexity Organs	23
4	Discussion and Conclusion	24
4.1	Tumour Geometric Complexity as a Geometric Descriptor: Interpretation and Limitations	24
4.2	Treatment Planning Across Modalities: What We Learned	25
4.3	Towards a Richer Quality-Assurance Framework: What the Formulation Enables	25
4.4	Overall Conclusions	26
5	Future Work	27
5.1	Scaling to the Full Patient Cohort	27
5.2	Validating the Four-Class Framework	27
	References	28
A	Skills and Knowledge Acquired	32

List of Figures

2.1	Distance-to-target histogram (DTH) of an organ at risk, partitioned into the overlap, near-compression, and separation zones. The overlap zone is the single bin at $r = 0$; the near-compression/separation boundary is the machine-derived threshold distance d_c (Eq. 2.2).	8
2.2	Proposed complexity-aware QA decision flow. The binary pass-or-fail verdict is extended to four classes by checking the TGC grade when only one of the two goals is met.	15
3.1	Distance-to-target histograms relative to the planning target volume for two organs at risk of the representative patient: the optic pathways ($d_c = 2.82$ mm, left) and the brain ($d_c = 1.76$ mm, right). The overlap, near-compression, and separation zones are shaded; the dashed line marks the critical distance d_c separating the near-compression and separation zones.	17
3.2	Dose distributions for the representative patient: the two-beam photon, proton, helium, and carbon plans (A–D). The photon plan shows the broadest spread of dose into the surrounding tissue, while the ion plans confine the dose more tightly around the target.	20

3.3	Dose–volume histograms (DVH) for the representative patient: two-beam photon, proton, helium, and carbon plans (A–D). All plans achieve full planning target volume coverage; the plans differ mainly in the dose carried by the organs at risk at intermediate dose levels.	21
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List of Tables

2.1	Interpretation of the organ-level Geometric Volume Coefficient, following Eq. (2.6). The two limits correspond to the engaged volume lying wholly in the overlap zone and wholly in the near-compression zone.	10
2.2	Planning parameters for the two-beam comparative plans (photons, protons, helium ions, and carbon ions).	13
3.1	Organ-level geometric decomposition and Geometric Volume Coefficient (GVC) for the ten organs at risk of the representative patient, computed relative to the PTV. Normalised Overlap Volume (NOV), Normalised Compression Volume (NCV), and Normalised Separation Volume (NSV) are the normalised volume fractions in the overlap, near-compression, and separation zones; $GVC = NOV + 0.5NCV$	17
3.2	Patient-level GVC calculation. Each organ’s squared complexity is weighted by its clinical priority w_{priority} ; the sum of the $w_{\text{priority}} \cdot GVC^2$ column is the patient-level GVC, and the last column gives each organ’s percentage contribution to it. The five organs-at-risk with $GVC = 0$ are omitted, as they do not contribute. . .	18
3.3	Organ-level Geometric Volume Coefficient (GVC) for the ten organs at risk of three head-region patients, computed relative to the PTV. The final row gives the patient-level GVC obtained by the quadratic aggregation of Eq. (2.8). Patient 1 is the representative case detailed in Sections 3.1.1–3.1.3.	19
3.4	Mean dose to the planning target volume (PTV) and organs at risk, relative to prescription, for the two-beam photon, proton, helium, and carbon plans of the three patients. Photon values are physical dose; all particle values are dose weighted by the relative biological effectiveness (RBE).	21
3.5	Maximum dose to the planning target volume (PTV) and organs at risk, relative to prescription, for the two-beam photon, proton, helium, and carbon plans of the three patients. Photon values are physical dose; all particle values are dose weighted by the relative biological effectiveness (RBE).	22

3.6	Spearman rank correlation coefficient between the organ-level Geometric Volume Coefficient (GVC) and the mean delivered dose, computed over all ten organs at risk, for each modality and each patient. In every case the correlation is positive, meaning that organs with a higher Geometric Volume Coefficient receive a higher mean dose, and all correlations are significant (one-tailed; ** $p < 0.01$).	22
3.7	Sensitivity analysis: Spearman rank correlation coefficient between the organ-level Geometric Volume Coefficient (GVC) and the mean delivered dose, computed over the non-zero-GVC organs at risk only ($n = 5$ for Patient 1, $n = 6$ for Patients 2 and 3). Significance (one-tailed) is marked * $p < 0.05$ and ** $p < 0.01$; the remaining values are not significant at $p < 0.05$	23

List of Abbreviations

Abbreviation	Full Form
3D-CRT	Three-Dimensional Conformal Radiotherapy
BED	Biologically Effective Dose
CT	Computed Tomography
CTV	Clinical Target Volume
DTH	Distance-to-Target Histogram
DICOM	Digital Imaging and Communications in Medicine
DVH	Dose–Volume Histogram
EBRT	External Beam Radiotherapy
GTV	Gross Tumour Volume
GVC	Geometric Volume Coefficient
IMRT	Intensity-Modulated Radiotherapy
LEM	Local Effect Model
LET	Linear Energy Transfer
LINAC	Linear Accelerator
NCV	Normalised Compression Volume
NOV	Normalised Overlap Volume
NSV	Normalised Separation Volume
OAR	Organ at Risk
PTV	Planning Target Volume
QA	Quality Assurance
RBE	Relative Biological Effectiveness
RT	Radiotherapy
RTDOSE	Radiotherapy Dose (DICOM object)
RTSTRUCT	Radiotherapy Structure Set (DICOM object)
SVDPB	Singular-Value-Decomposed Pencil Beam
TGC	Tumour Geometric Complexity
VMAT	Volumetric Modulated Arc Therapy

Chapter 1

Introduction

1.1 Historical Development of Radiotherapy

Radiotherapy came out of physics. Wilhelm Röntgen discovered X-rays in 1895, radioactive sources were identified soon after, and controllable beams of ionising radiation became available for medical use [2, 3]. Early treatments reached only superficial disease, because the photon energies then available gave high entrance doses and deposited little at depth. The discovery of radium opened a second route, brachytherapy, in which sources placed in or near the target raise the local dose while sparing the surrounding tissue [3].

After World War II, advances in nuclear physics and particle acceleration carried the field into the megavoltage era. Cobalt-60 teletherapy came first, with much improved depth dose and skin sparing, and it standardised dosimetry and calibration [3]. Medical linear accelerators (LINACs) then took over, giving higher dose rates, sharper penumbra, and selectable energies. Intensity-modulated radiotherapy (IMRT) and volumetric modulated arc therapy (VMAT) now produce highly conformal dose distributions. A parallel line of development left photons altogether: following Wilson’s proposal that the finite range and Bragg peak of charged particles could confine dose more tightly than any photon beam, proton and heavier-ion therapy entered clinical use [19].

Neither conformality nor a sharper fall-off has removed normal-tissue complications. Even when the planned dose is delivered accurately, whether a plan can spare the organs at risk depends heavily on the patient’s anatomy: on how the target is shaped, and on how closely it sits to the critical structures. The intrinsic difficulty of a case is therefore central to the quality of radiotherapy, and it needs to be measured reproducibly. That measurement is the subject of this thesis.

1.2 Principles of Radiotherapy

Radiotherapy (RT) is the clinical application of ionising radiation to deliver a prescribed dose to a target volume while limiting deposition in the surrounding normal tissue. It can usefully be framed as an inverse problem: given the clinical objectives and constraints, find a deliverable irradiation configuration whose 3D dose field optimises the tumour–normal-tissue trade-off. Advances in imaging, dose calculation, and delivery have made steep dose gradients possible, but

how far a gradient can spare an organ is constrained by where that organ lies relative to the target. It is this geometric constraint that the present work sets out to quantify. Approximately 50–60% of all cancer patients receive radiotherapy at some stage of their treatment [1].

1.2.1 Radiotherapy Techniques

External beam radiotherapy (EBRT) is the most widely used modality, and the progression from three-dimensional conformal radiotherapy (3D-CRT) to IMRT and VMAT reflects a steady increase in the degrees of freedom available for shaping dose. These techniques matter most in head-and-neck treatments, where the target volumes lie close to radiosensitive structures. Small changes in beam modulation shift the dose gradients across the organs at risk, and an acceptable plan usually requires balancing competing constraints over many structures at once. This interplay between geometry, delivery, and dose gradients determines whether a high-quality plan is achievable at all.

Delivery has continued to grow more sophisticated. Particle therapy gives steeper dose gradients than photons; image-guided and adaptive techniques allow the plan to be re-optimised to the anatomy of the day; and knowledge-based planning is beginning to make the planning step itself algorithmic. Each of these raises either the number of plans requiring review or the difficulty of judging any one of them by eye. The sophistication of delivery has outpaced that of quality assurance. And however sharp the achievable gradient becomes, it cannot fully spare an organ that lies inside the target. Part of the difficulty is anatomical, and it is fixed before any beam is chosen.

1.3 Clinical Trials and the Role of Quality Assurance

New cancer treatments are evaluated in clinical trials before they enter routine practice. Early-phase studies establish whether a treatment is safe; later phase III trials compare it against the current standard of care in large patient populations, usually across many institutions. Such trials provide the highest level of clinical evidence, and their findings determine how patients are treated in practice.

Radiotherapy poses a particular difficulty in this setting. A pharmaceutical agent is standardised during manufacture and is therefore identical for every patient and every centre. Radiotherapy is prepared individually, for each patient, at the treating institution. A clinician delineates the tumour and the organs at risk on the patient’s imaging, a planner designs the dose distribution, and the treatment machine delivers it. Each of these steps may differ from one centre to another, so a treatment that is nominally the same across a trial can vary considerably between institutions.

Quality assurance (QA) exists to limit this variability. It is the set of checks verifying that each plan conforms to the trial protocol. Before an institution may enrol patients it must pass credentialing checks, and during the trial individual plans are reviewed against the protocol.

These checks are not administrative formalities, because the quality of radiotherapy can determine whether patients survive. In the large head-and-neck trial TROG 02.02, patients whose plans contained serious deficiencies fared markedly worse: about 50% were alive at two

years, against 70% of those whose plans complied with the protocol from the outset [5]. A later meta-analysis of multiple trials found the same pattern, with protocol deviations associated with roughly a 1.7-fold higher risk of death and an increased risk of treatment failure [6]. Rigorous QA therefore protects both the individual patient and the integrity of the trial [7].

Despite this evidence, QA remains unevenly applied. A review of phase III radiotherapy trials found that only about two-thirds required any form of QA, that fewer than half mandated prior credentialing of the participating centres, and that trials using the newest and most complex techniques were no more likely to include QA than older ones [8]. Treatments have grown more complex and QA has not kept pace. To see why this matters, it helps to look closely at how QA is actually carried out.

1.4 The Current QA Process and Its Limitations

A typical example makes the process concrete. In a multicentre phase III head-and-neck trial delivering IMRT to about 70 Gy, a small group of expert reviewers, usually about three, check each plan against the protocol before treatment. They assess how the tumour and the organs were delineated, the prescribed dose, the coverage of the tumour, and the dose reaching the healthy organs. Most plans pass, only a small fraction are flagged, and the protocol rules are applied with some clinical flexibility. Three limitations follow from working this way.

First, the process depends on people. Experts do not always delineate organs in the same way, and they do not always judge plans in the same way, so the outcome carries a degree of human variability [11]. Expert review is also slow and hard to scale: as accrual grows, reviewing every plan by hand becomes a heavy burden.

Second, the review yields little more than a pass-or-fail verdict. When a plan is flagged as fail, the verdict does not say whether the failure came from sub-optimal planning or from an anatomy too complex for the goals to be met. Nor does it indicate how the patient is likely to fare, which is what clinicians most need to know: whether the tumour will be controlled, whether the patient will survive, and how severe the side effects will be when graded on a standard scale [4]. There is growing interest in moving QA from asking whether a plan follows the rules towards asking what outcome it is likely to produce, and the evidence points that way, since in one trial the expert review of complex cases was found to relate to patient survival [9]. This does not yet form part of the workflow.

Third, the review considers the plan largely in isolation. The patient's clinical history, prior treatment, comorbidities, and individual risk factors are most often not taken into account, even though they bear directly on how much dose an organ can tolerate and on the outcome that is ultimately of interest.

The current QA process is therefore manual, partly subjective, restricted to a pass-or-fail outcome, and disconnected from both prognosis and the individual patient's clinical context. These are not shortcomings of the people who perform it. They are the inherent limits of a qualitative, manual process, and they point directly to where improvement is possible.

1.5 Areas of Improvement

The limitations described above point to two directions for improvement.

The first is to make QA more transparent and standardised by grading how difficult each case is. Current QA quantifies case difficulty nowhere, yet it is that difficulty which determines whether a high-quality plan is achievable at all. Express the complexity of a case as a number and several benefits follow at once: the binary pass-or-fail verdict can be opened into finer classes that distinguish a plan limited by poor planning from one limited by the anatomy, review can be concentrated on the hardest cases, and hospitals and techniques can be compared on an equal footing. This number is what the present work terms *Tumour Geometric Complexity* (TGC), and building it is the central aim of the thesis. The concept of quantifying case difficulty through distance-based measures was introduced by Özer et al. [18], who demonstrated its feasibility in the context of photon-based head-and-neck radiotherapy using non-standard planning structures. Caen is now moving beyond photons: the CYCLHAD centre, with the Centre François Baclesse, has treated patients with proton therapy since 2018 and is preparing a multi-ion platform for carbon-ion beams. A difficulty measure built for photons alone would risk obsolescence as the centre moves to ions. The present descriptor therefore develops that concept into an organ- and patient-level measure tested across four delivery modalities instead of one.

The second direction is automation. Once case difficulty is quantified and the QA outcome expressed through these richer classes, the classification itself can be automated, so that review becomes faster and more consistent even at high volume. This would not replace the experts but support them, releasing their time for the genuinely difficult cases. It also opens the way to linking QA to outcomes, since a reproducible complexity grade is exactly the kind of feature that can be related to tumour control, survival, and toxicity.

This thesis concentrates on the first direction while setting up the second, because a sound complexity measure is the foundation the automation builds on.

1.6 Treatment Planning and Anatomical Complexity

Planning is largely a matter of compromise. Many objectives must be satisfied at once and they often conflict, so improving coverage of the tumour may raise the dose in a neighbouring organ. How far the compromise can be pushed depends on the patient’s geometry. When an organ lies close to the target, or inside it, re-optimising the plan again and again may still not bring it within the desired goals, because the difficulty is anatomical and not a matter of effort. When the same organ is well separated, the goals are usually easy to meet. This dependence is greatest in head-and-neck cancer, where the tumour and nodal regions often form a large, curved, branching structure among many sensitive organs, and where surgery may leave the anatomy more irregular still.

The modality also affects the picture, since photon IMRT, proton therapy, and other techniques accommodate complex shapes differently, so the same anatomy can be easier for one modality than for another. The present work examines this in matRad, an open-source treatment-planning system able to generate and compare plans across modalities for the same patient [12].

The proximity of the organs at risk to the target, and the degree to which the target encloses them, can be expressed numerically and combined into a single reproducible measure. This is the Tumour Geometric Complexity that the thesis sets out to develop.

1.7 Objective of the Thesis

This thesis sets out to establish a reproducible geometric measure of how intrinsically difficult a radiotherapy case is, and to show that this measure reflects the dose burden the anatomy imposes on the organs at risk. Meeting this aim requires two preparatory steps, each a contribution in its own right:

- Formulating and computing the measure. Tumour Geometric Complexity is defined through the Geometric Volume Coefficient, a descriptor built from the delineated contours alone that combines the position of each organ at risk relative to the target with the limit of machine deliverability. It is computed for the individual organ, which localises the difficulty, and aggregated to the patient, which grades the case as a whole (Section 3.1).
- Obtaining the delivered dose across modalities. Plans are generated in matRad for photons, protons, helium ions, and carbon ions on the same patients and the same beam geometry, to record the dose received by every structure under each modality (Section 3.2) [12].

The two are then brought together. The organ-level coefficient, computed from the contours before any plan exists, is compared against the mean dose those same organs actually receive (Section 3.3). If the geometric ordering reproduces the dosimetric ordering, and does so for every modality and every patient, the descriptor measures intrinsic difficulty and not the artefact of one particular optimisation. Section 2.4 sets out the four-class QA scheme that such a grade makes possible; validating it, scaling the computation to the full cohort, and automating the classification are future work (Chapter 5).

Chapter 2

Methods

Current quality-assurance review returns a binary verdict, and on a failure it gives no way to tell a badly made plan from an anatomy that never allowed a good one. This chapter builds the measure that separates the two, in four steps. Section 2.1 formulates Tumour Geometric Complexity through the Geometric Volume Coefficient, at the organ and patient level, from the geometry alone. Section 2.2 describes the multi-modality planning study in matRad, which gives the dose actually delivered to each organ under each modality. Section 2.3 sets out how the agreement between the two is assessed. Section 2.4 proposes how the measure turns the binary verdict into a four-class scheme.

2.1 Tumour Geometric Complexity: Formulation and Computation

The Geometric Volume Coefficient developed in this section is motivated by two physical facts. Organ volume lying inside the target cannot be spared by any plan, and volume lying closer to the target than the achievable dose gradient allows cannot be spared by a real machine. The descriptor separates these two contributions and penalises them accordingly. It is not derived from a variational principle or a conservation law; its form is proposed from physical and clinical reasoning. The proposal is kept separate from its test: this chapter constructs the descriptor, and Chapter 3 examines, on three cases, whether it tracks the delivered dose.

Tumour Geometric Complexity expresses how intrinsically difficult a case is, on the basis of how the organs at risk are arranged around the target. The descriptor that quantifies it is the Geometric Volume Coefficient, defined first for a single organ and then aggregated to the whole patient. The two levels answer different questions and are kept distinct for that reason. The organ-level coefficient is diagnostic: it identifies *which* structures drive the difficulty of a case, and in the QA scheme of Section 2.4 it is what explains a verdict. The patient-level value is a single grade for the whole case, the number read at the branch point of the four-class decision. One localises the difficulty; the other summarises it.

The coefficient uses only the contours and requires no dose calculation. Its overlap term is purely anatomical, while its near-compression term takes in machine deliverability through the achievable dose gradient. The descriptor is machine-aware in this single respect, and that is

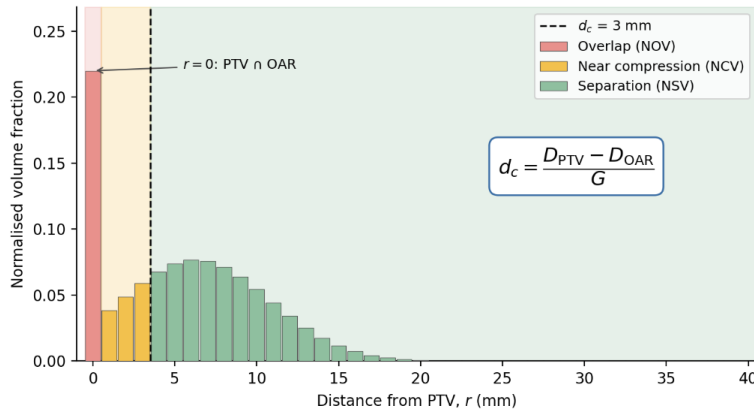
what allows it to be applied consistently across delivery modalities.

2.1.1 Organ-level Geometric Volume Coefficient

The spatial relationship between an organ at risk and the target has been represented since Kazhdan et al. [20] by the overlap volume histogram, and equivalently by the distance-to-target histogram of Zhu et al. [21], which records the fraction of organ volume as a function of its distance from the target. Yuan et al. [22] showed, on 64 prostate and 82 head-and-neck plans, that the fraction of an organ lying within a given distance range of the target is a significant predictor of the dose it receives. That range, however, was fitted empirically from a training database and came out different for every organ. The present work introduces a three-zone decomposition of the histogram whose boundary is instead set analytically from machine deliverability, so that no training is required and the extension to other modalities is immediate.

The descriptor is defined on the voxel grid, since that is how it is computed. Every voxel of an organ at risk is assigned its Euclidean distance r to the nearest voxel of the planning target volume, and voxels lying inside the target take $r = 0$. The $r = 0$ bin is therefore exactly the geometric intersection $PTV \cap OAR$, and the near-compression zone begins at the first bin beyond it. The two boundaries $r = 0$ and $r = d_c$ partition the organ into three zones (Figure 2.1): an overlap zone, which cannot be spared and represents intrinsic complexity; a near-compression zone, too close to the target to be spared given the achievable dose gradient, which represents the machine limitation; and a separation zone, far enough from the target to be spared and effectively free of risk. Counting the voxels in each zone gives three dimensionless fractions, the Normalised Overlap Volume (NOV), the Normalised Compression Volume (NCV), and the Normalised Separation Volume (NSV):

Figure 2.1: Distance-to-target histogram (DTH) of an organ at risk, partitioned into the overlap, near-compression, and separation zones. The overlap zone is the single bin at $r = 0$; the near-compression/separation boundary is the machine-derived threshold distance d_c (Eq. 2.2).



$$NOV = \frac{N_{r=0}}{N_{OAR}}, \quad NCV = \frac{N_{0 < r \leq d_c}}{N_{OAR}}, \quad NSV = \frac{N_{r > d_c}}{N_{OAR}}, \quad (2.1)$$

where each N counts the organ voxels falling in the zone indicated and N_{OAR} is the total number of organ voxels. Each voxel has a single distance and falls in exactly one zone, so the three counts

sum to N_{OAR} and the three fractions tile the organ, $\text{NOV} + \text{NCV} + \text{NSV} = 1$. All three are counting fractions, and the overlap fraction is the count at a single bin rather than over a range. A signed-distance convention, which would spread the overlap across negative distances as in the histogram literature, is noted as a refinement in Section 5.1.

The threshold d_c is set by machine deliverability. It is the minimum distance over which the dose can fall from the target prescription D_{PTV} to the organ tolerance D_{OAR} at the steepest achievable dose gradient G ,

$$d_c = \frac{D_{\text{PTV}} - D_{\text{OAR}}}{G}, \quad (2.2)$$

which follows the distance-based complexity concept of Özer et al. [18]; the three-zone decomposition and the coefficient developed below give that principle an explicit form and extend it. In clinical practice G is obtained through machine commissioning; here it was derived from the matRad dose distribution (Section 2.1.4). The machine constraint therefore enters only through the position of the boundary: organ volume lying closer than d_c cannot be spared even by an ideal plan. The overlap term carries no dependence on G . The part of an organ inside the target is irreducible whatever the delivery technique, and it forms a lower bound the coefficient cannot fall below.

The organ-level Geometric Volume Coefficient is a penalised sum of the three fractions,

$$\text{GVC} = \alpha \text{NOV} + \beta \text{NCV} + \gamma \text{NSV} \quad (2.3)$$

with the penalties ordered $\alpha \geq \beta \geq \gamma \geq 0$, the overlap zone penalised most and the separation zone least. This ordering is the only constraint the formulation imposes; the simplest instance satisfying it is $\alpha = 1$, $\beta = 0.5$, $\gamma = 0$, used throughout, for which the separation term drops out,

$$\text{GVC} = \text{NOV} + \frac{1}{2} \text{NCV}. \quad (2.4)$$

These penalties are fixed rather than fitted. Setting $\alpha = 1$ and $\gamma = 0$ fixes the scale and confines the coefficient to $[0, 1]$: the overlap zone counts in full, the separation zone not at all. The ordering then leaves β free in $[0, 1]$, and 0.5 is its midpoint, taken as a first-hand choice because the near-compression zone is intermediate in kind, neither irrecoverable like the overlap zone nor free like the separation zone. One unit of near-compression volume therefore counts as half a unit of overlap. The choice adds no parameter to be tuned and keeps the coefficient interpretable; optimising it against dose on a larger cohort is left to future work (Section 5.1).

2.1.2 Interpretation of the organ-level coefficient

The coefficient weights the overlap zone at full value and the near-compression zone at half, so a single value does not fix the geometry that produced it. What it does fix is how much of the organ is not free of risk. Writing that engaged fraction as E ,

$$E = \text{NOV} + \text{NCV} = 1 - \text{NSV}, \quad (2.5)$$

the two extremes bracket it. If the engaged volume lies wholly inside the target, the coefficient equals it directly. If instead it lies wholly in the near-compression band, only half of it is counted, so twice as much volume is needed to reach the same coefficient. Every intermediate geometry falls between these two limits, and none lies outside them, since no zone fraction can be negative and the engaged volume cannot exceed the whole organ. Hence

$$\text{GVC} \leq E \leq \min(2 \text{GVC}, 1). \quad (2.6)$$

The coefficient is therefore a lower bound on the engaged fraction and half an upper bound on it, locating it to within a factor of two. The two bounds coincide when the organ lies wholly in the separation zone, and again when it lies wholly inside the target.

Table 2.1: Interpretation of the organ-level Geometric Volume Coefficient, following Eq. (2.6). The two limits correspond to the engaged volume lying wholly in the overlap zone and wholly in the near-compression zone.

Geometric Volume Coefficient	Engaged Volume (%)	Safe Volume (%)
0.0	0	100
0.2	20–40	60–80
0.5	50–100	0–50
1.0	100	0

The coefficient thus states how much of an organ was never available to be spared. A value near zero means that any dose the organ receives is a property of the plan; a value near unity means that no re-optimisation will remove it. No threshold separating a complex anatomy from a simple one is proposed here, as fixing that operating point requires a cohort with recorded quality-assurance outcomes (Section 5.2). Where a single value is ambiguous, the three zone fractions are reported for every organ (Table 3.1) and resolve it.

2.1.3 Patient-level Coefficient

Organs contribute unequally to the difficulty of a case, both because their complexities differ and because their clinical importance differs. The patient-level measure must therefore let the most complex organs dominate without discarding the contribution of the less complex ones, and it must weight organs by clinical importance in a standardised way. Two weights are needed. Beginning from a general weighted sum over the N organs of a case,

$$\text{GVC}_{\text{patient}} = \sum_{i=1}^N w_{\text{rank},i} w_{\text{priority},i} \text{GVC}_i, \quad (2.7)$$

the requirement that more complex organs count for more is met, without introducing a further free parameter, by setting each organ’s rank weight equal to its own complexity, $w_{\text{rank},i} = \text{GVC}_i$.

This self-weighting turns the sum into a quadratic aggregation,

$$\text{GVC}_{\text{patient}} = \sum_{i=1}^N w_{\text{priority},i} \text{GVC}_i^2 \quad (2.8)$$

with $\text{GVC}_i \in [0, 1]$ and $w_{\text{priority},i} \in [0, 1]$. The quadratic form is chosen for three reasons. It emphasises the clinically critical structures, since a single highly complex organ then dominates the score, which is the behaviour a case grade should have: a patient with one inseparable organ is a difficult case whatever the rest of the anatomy looks like. It avoids the dilution that an average would produce, because organs with a small coefficient still contribute their share instead of being cancelled by the well-separated ones. And it introduces no exponent to be tuned, in the manner of the quadratic, energy-like aggregations common in physics. The result is an unbounded weighted sum. It may exceed unity when several high-priority organs are strongly engaged with the target, and it is therefore read as a relative severity score that discriminates between geometries, not as a normalised coefficient.

The clinical priority weight w_{priority} encodes the severity and irreversibility of the toxicity associated with irradiating a structure, highest for serial, vision- and life-critical organs such as the optic apparatus and brainstem, lower for the brain and hippocampus. The weights are a modelling choice, as no standardised institutional weighting scheme exists, but they are not set subjectively: they are based on current clinical knowledge, drawn from established normal-tissue tolerance and protocol sources so that the same organ receives the same weight at every centre [14, 13, 15], and they can be refined against the institutional (Baclesse) protocols and as larger datasets become available.

2.1.4 Computation

Both quantities were computed from the delineated structures of the planned patients, the same structures later used for planning (Section 2.2.1). The planning target volume was masked and merged with each organ mask, and for every organ voxel the Euclidean distance to the nearest target voxel was computed, which gives the counts of Eq. (2.1) directly. The computation uses only the contours, so the coefficient carries no dose information of its own; its sole machine-dependent input is the gradient G that fixes d_c . The threshold (1.76–2.82 mm for the organs reported here) generally falls within a distance bin and not on a boundary. Each bin was assigned to the zone containing its centre, and the resulting fractions were stable to a one-bin shift for the organs carrying non-zero complexity.

The gradient G was not assumed but obtained from the matRad dose distribution. It was evaluated across the PTV boundary and taken as the 95th percentile of that distribution rather than its maximum, since the maximum can reflect a single noisy voxel and not a representative achievable fall-off. This gave $G = 5.67$ Gy/mm, which fixes the threshold distance used throughout.

What remains is to ask whether a measure computed from the contours alone reflects the dose an organ actually receives. Answering that requires the delivered doses, which the following planning study provides. It spans four modalities for a reason internal to the descriptor’s claim: if

the coefficient measures difficulty intrinsic to the anatomy, it should track dose regardless of how the dose is delivered, and a single-modality check cannot test this. The same requirement makes the descriptor relevant to the arrival of hadron therapy in Caen, where a measure developed for one delivery technique would risk obsolescence as the institution moves to ions.

2.2 Multi-Modality Treatment Planning with matRad

2.2.1 Patient data and structures

The study used data from three previously treated patients with tumours in the head region, selected for anatomies in which the target lies near critical structures, in particular the optic nerves and the optic chiasm. The cases were drawn from a bi-institutional, review-board-approved study (approvals CFB CSE INT-24-016 and University Hospital CLERS ID 5496, 5 April 2024), conducted in compliance with the Declaration of Helsinki and the GDPR. All data were anonymised, and the cases were handled according to the applicable ethical and data-protection requirements. Each case was provided in DICOM format and comprised the planning CT, which carries the anatomical information and the tumour location, and the RTSTRUCT file, which holds the delineated target volumes (GTV, CTV, PTV) and the organs at risk (OARs). Extension to the full cohort is left to future work (Chapter 5).

The head region was chosen because the target lies among several radiosensitive structures: the optic chiasm, the optic nerves, and the retinas, among others. In such anatomy small dosimetric differences can have large clinical consequences, which makes it a demanding and informative test case.

2.2.2 The planning workflow

All plans were produced in matRad, an open-source toolkit for dose calculation and plan optimisation that operates in a manner comparable to a clinical treatment-planning system [12]. matRad was chosen for its flexibility and because a commercial planning system was not available for this work.

The workflow had four steps: importing the planning CT and segmenting the structures from the RTSTRUCT contours; configuring the beams by selecting the modality (photons, protons, helium ions, or carbon ions) and the gantry and couch angles; computing and optimising the dose against the prescribed objectives; and analysing the result through dose–volume histograms (DVHs), from which the mean and maximum dose to each structure were extracted. These per-structure dose statistics are the planning endpoints, and they are what Chapter 3 tests the geometric measure against.

2.2.3 Plan configurations and modalities

Four plans were generated, using a common two-beam arrangement for all modalities so that photons, protons, helium ions, and carbon ions could be compared on an equal geometric footing. The shared parameters are listed in Table 2.2.

Table 2.2: Planning parameters for the two-beam comparative plans (photons, protons, helium ions, and carbon ions).

Parameter	2-beam (all modalities)
Number of beams	2
Prescribed dose	30 Gy
Number of fractions	30
Bixel width	2.5 mm
Dose grid resolution	$2.5 \times 2.5 \times 2.5$ mm
Gantry angles	350, 190°
Couch angles	0, 0°
Iso-centre	(0.0, 3.90, -7.40)
PTV objective	squared deviation ($w = 300$, $d_{\text{ref}} = 30$)

The prescription of Table 2.2 is not a clinical schedule. A nominal 30 Gy in 30 fractions was adopted as a common normalisation for the four plans, so that every modality is optimised to the same target dose and the organ-at-risk doses can be reported relative to prescription and compared directly. The comparison of Section 3.3 rests on the ranking of the organs within each plan, so the absolute prescription level does not affect it, and the fixed number of fractions is what allows the biological modelling to be handled as described below. The fine bixel width and matching grid resolution give good target conformity, which matters most for carbon ions.

All plans used matRad’s Generic machine base data (version “Cleve” v3.1.0). Dose was computed with the analytical singular-value-decomposed pencil-beam engine (SVDPB), matRad’s default for rapid dose calculation and optimisation [12]. matRad also provides Monte Carlo engines, ompMC for photons and a TOPAS/Geant4 interface for particles, which model transport through heterogeneous tissue such as bone and air cavities more accurately, at substantially higher computational cost [12].

These plans are simplified relative to a clinically validated treatment plan. They were produced in a research planning system on a fixed beam arrangement, with a single objective on the target, and were not subject to the iterative refinement and clinical review that a treating centre applies. The simplification is deliberate, since the study serves as a controlled dosimetric reference for the geometric descriptor and not as a clinical planning exercise, but the resulting doses should not be read as those a treating centre would deliver.

Biological modelling and the optimised quantity. matRad can optimise and report a plan on several quantities of increasing biological content: physical dose D ; the biological *effect* $\varepsilon = \alpha D + \beta D^2$, the linear–quadratic quantity from which cell survival is derived; the RBE-weighted dose D_{RBE} , obtained from the effect and expressed on the photon-equivalent scale; and the biologically effective dose $\text{BED} = D (1 + d/(\alpha/\beta))$, which additionally accounts for fractionation through the tissue α/β ratio [12, 16]. Here the plans were optimised and evaluated on RBE-weighted dose. All four share a single fractionation schedule (30 fractions), so RBE-weighted dose already places the modalities on a common biological footing. The effect- and BED-based quantities require tissue-specific α/β values and become relevant chiefly when fractionation

differs; they were therefore not needed here.

Each modality was configured with its standard biological model. The photon plan was optimised and evaluated on physical dose, for which the relative biological effectiveness (RBE) is taken as unity. The particle plans were optimised and evaluated as RBE-weighted dose: a constant RBE of 1.1 was applied for protons [16]; the Local Effect Model (LEM) was used for carbon ions [10]; and a data-driven parameterisation based on the dose-averaged linear energy transfer (LET) was used for helium ions [17]. All plans therefore share a common effective-dose scale. The three particle models are not mutually calibrated, so a strict ion-versus-ion dosimetric comparison would require a common biological basis. This is stated as a limitation in Section 4.2; it does not affect the ranking-based analysis on which the descriptor is tested.

To keep the comparison between dose deposition and geometric complexity unbiased, no planning objectives or dose-volume constraints were applied to the organs at risk during inverse optimisation. The optimisation was driven solely by the PTV objective of Table 2.2. The dose received by each organ at risk was therefore determined purely by its spatial proximity to the PTV and by the physical delivery characteristics of each modality, so that the subsequent correlation analysis reflects the unmanipulated geometric-dosimetric relationship.

2.3 Agreement Between Geometric Volume Coefficient and Dosimetric Outcomes

The geometric measure is meaningful only if it reflects the dose an organ actually receives. The agreement was therefore assessed on each patient by placing every organ’s contour-only coefficient alongside its mean delivered dose and comparing the two rankings by the Spearman rank correlation, for each modality separately. If the coefficient captures intrinsic difficulty, the ordering should hold across all four, since it is the anatomy and not the delivery technique that the measure describes. A number of organs lie entirely in the separation zone and take a coefficient of exactly zero, so the comparison was made twice, once over all organs and once over the non-zero organs alone, in order to make visible the contribution of these tied values. The quantitative comparison is reported in Chapter 3.

2.4 Proposed Integration of Tumour Geometric Complexity into the QA Framework

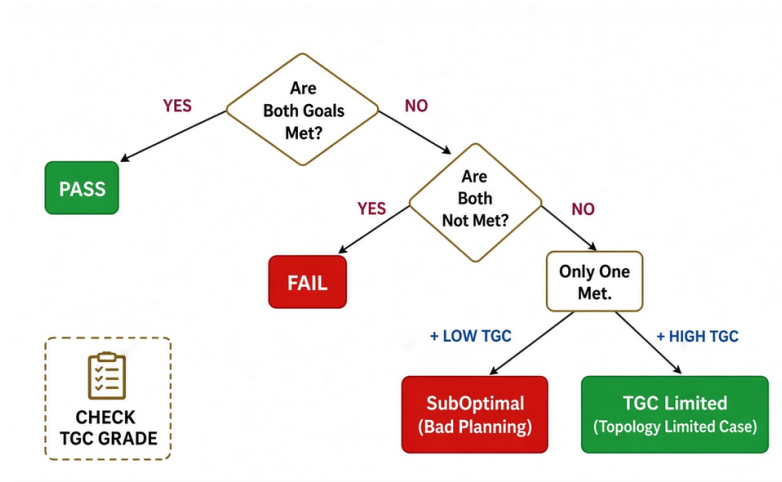
Current QA reaches a binary verdict and cannot separate a poorly made plan from an impossible one. Tumour Geometric Complexity supplies the information needed to separate them, and this section proposes how.

A plan is assessed against two goals: adequate target (PTV) coverage and adequate sparing of the organs at risk. Combining these two outcomes with the case complexity gives four classes instead of two. A plan **passes** when both goals are met and **fails** when neither is. When exactly one goal is met, the complexity grade decides. A low-complexity case is **sub-optimal**: the anatomy was not the obstacle, so the shortfall reflects poor planning and the plan should be

reworked. A high-complexity case is **complexity-limited**: the anatomy forced a compromise no plan could avoid, so the plan is theoretically failing but clinically acceptable.

The decision logic is shown in Figure 2.2. The organ-level coefficient indicates which structures drive the difficulty, while the patient-level score provides the grade at the branch point, so that the verdict rests on a reproducible measure and not on expert judgement alone.

Figure 2.2: Proposed complexity-aware QA decision flow. The binary pass-or-fail verdict is extended to four classes by checking the TGC grade when only one of the two goals is met.



This four-class scheme is the proposed end point of the thesis. It is presented as a framework and is not validated on patient data here; its support is the geometry–dose agreement of Chapter 3. Validating the four classes against clinical outcome, and automating the classification, are future work (Chapter 5).

Chapter 3

Results

The full pipeline is carried through in detail for a single representative head-region patient, and the organ-level geometric result is then extended to three patients. Section 3.1 reports the Tumour Geometric Complexity at the organ and patient level; Section 3.2 reports the matRad planning study; and Section 3.3 closes with the agreement between the geometric measure and the delivered dose, the central result of the thesis.

3.1 Tumour Geometric Complexity: Organ-Level and Patient-Level Computation

3.1.1 Organ-level Geometric Volume Coefficient

The Geometric Volume Coefficient was computed for each of the ten organs at risk delineated in the planning study, from the patient contours relative to the planning target volume, following Section 2.1, with $\alpha = 1$, $\beta = 0.5$, $\gamma = 0$. The values are given in Table 3.1.

Five structures carry non-zero complexity. The optic chiasm is highest, with over 80% of its volume inside the overlap and near-compression zones. The left hippocampus follows by a different route: almost none of it overlaps the target, yet three-quarters lies in the near-compression band, which drives its coefficient to the second-highest value. The optic pathways, brainstem, and brain lie progressively further into the separation zone, and the remaining five structures return exactly zero.

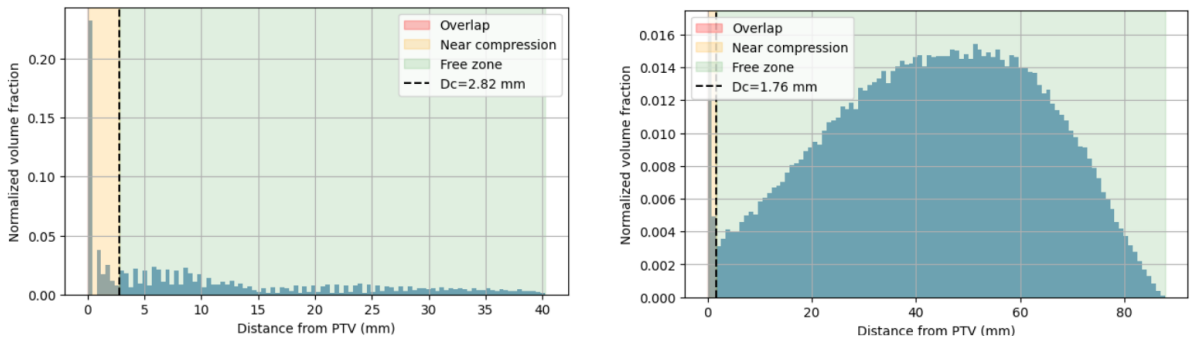
Table 3.1: Organ-level geometric decomposition and Geometric Volume Coefficient (GVC) for the ten organs at risk of the representative patient, computed relative to the PTV. Normalised Overlap Volume (NOV), Normalised Compression Volume (NCV), and Normalised Separation Volume (NSV) are the normalised volume fractions in the overlap, near-compression, and separation zones; $GVC = NOV + 0.5 NCV$.

Organ	NOV	NCV	NSV	GVC
Chiasma (optic chiasm)	0.4167	0.4063	0.1771	0.6198
Hippocampe G (hippocampus, left)	0.0038	0.7664	0.2298	0.3870
Voies Optiques (optic pathways)	0.2321	0.0991	0.6689	0.2816
Tronc (brainstem)	0.0638	0.0782	0.8580	0.1029
Encéphale (brain)	0.0166	0.0049	0.9784	0.0191
Hippocampe D (hippocampus, right)	0.0000	0.0000	1.0000	0.0000
Rétine G (retina, left)	0.0000	0.0000	1.0000	0.0000
Rétine D (retina, right)	0.0000	0.0000	1.0000	0.0000
Parotide G (parotid, left)	0.0000	0.0000	1.0000	0.0000
Parotide D (parotid, right)	0.0000	0.0000	1.0000	0.0000

3.1.2 Distance-to-target histograms

The distance-to-target histogram makes the decomposition of Table 3.1 explicit. Figure 3.1 shows two contrasting cases. The optic pathways retain a substantial volume fraction close to the target, with a visible mass inside the near-compression band bounded by $d_c = 2.82$ mm, consistent with the intermediate coefficient. The brain is almost entirely displaced into the separation zone: its distribution peaks far from the target and only a negligible fraction falls within $d_c = 1.76$ mm. The two cases bracket the descriptor’s range.

Figure 3.1: Distance-to-target histograms relative to the planning target volume for two organs at risk of the representative patient: the optic pathways ($d_c = 2.82$ mm, left) and the brain ($d_c = 1.76$ mm, right). The overlap, near-compression, and separation zones are shaded; the dashed line marks the critical distance d_c separating the near-compression and separation zones.



3.1.3 Patient-level Coefficient

The patient-level value was obtained through the quadratic aggregation of Eq. (2.8). Table 3.2 shows the calculation explicitly, so that the role of the two weights is visible: each organ’s complexity is squared, then multiplied by its clinical priority weight w_{priority} , and the products

are summed. The five structures with $GVC = 0$ contribute nothing and are omitted.

Table 3.2: Patient-level GVC calculation. Each organ’s squared complexity is weighted by its clinical priority w_{priority} ; the sum of the $w_{\text{priority}} \cdot GVC^2$ column is the patient-level GVC, and the last column gives each organ’s percentage contribution to it. The five organs-at-risk with $GVC = 0$ are omitted, as they do not contribute.

Organ	GVC	w_{priority}	GVC^2	$w_{\text{priority}} \cdot GVC^2$	Contribution (%)
Chiasma	0.6198	0.90	0.3841	0.3457	65.88
Hippocampe G	0.3870	0.65	0.1498	0.0974	18.55
Voies Optiques	0.2816	0.90	0.0793	0.0714	13.60
Tronc	0.1029	0.95	0.0106	0.0101	1.92
Encéphale	0.0191	0.75	0.0004	0.0003	0.05
Patient-level GVC = $\sum$				0.5248	100

The patient-level Geometric Volume Coefficient is 0.5248. More than 98% of it is carried by the three most entangled structures, the optic chiasm (65.9%), the left hippocampus (18.6%), and the optic pathways (13.6%), while the brainstem and brain supply the rest instead of being discarded.

3.1.4 Organ-level Coefficient Across Three Patients

To show that the descriptor behaves consistently beyond the single detailed case, the organ-level coefficient was computed for two further head-region patients (Table 3.3). The geometry differs from patient to patient, and so does the set of structures carrying non-zero complexity. In every case the optic chiasm is the most entangled, but the second-ranked organ changes: the left hippocampus for Patient 1, the optic pathways for Patient 2, and the right hippocampus for Patient 3. The descriptor responds to each patient’s specific anatomy and does not simply flag the same structure every time.

The patient-level values span a wide range, from 0.0406 for the most favourably separated case to 1.1509 for the most entangled. The score exceeds unity for Patient 2, as the unbounded aggregation of Eq. (2.8) permits when several high-priority structures are strongly engaged.

Table 3.3: Organ-level Geometric Volume Coefficient (GVC) for the ten organs at risk of three head-region patients, computed relative to the PTV. The final row gives the patient-level GVC obtained by the quadratic aggregation of Eq. (2.8). Patient 1 is the representative case detailed in Sections 3.1.1–3.1.3.

Organ	Patient 1	Patient 2	Patient 3
Chiasma (optic chiasm)	0.6198	0.9938	0.1395
Voies Optiques (optic pathways)	0.2816	0.5237	0.0938
Hippocampe G (hippocampus, left)	0.3870	0.1441	0.0099
Hippocampe D (hippocampus, right)	0.0000	0.0272	0.0970
Tronc (brainstem)	0.1029	0.0319	0.0913
Encéphale (brain)	0.0191	0.0187	0.0375
Rétine G (retina, left)	0.0000	0.0000	0.0000
Rétine D (retina, right)	0.0000	0.0000	0.0000
Parotide G (parotid, left)	0.0000	0.0000	0.0000
Parotide D (parotid, right)	0.0000	0.0000	0.0000
Patient-level GVC	0.5248	1.1509	0.0406

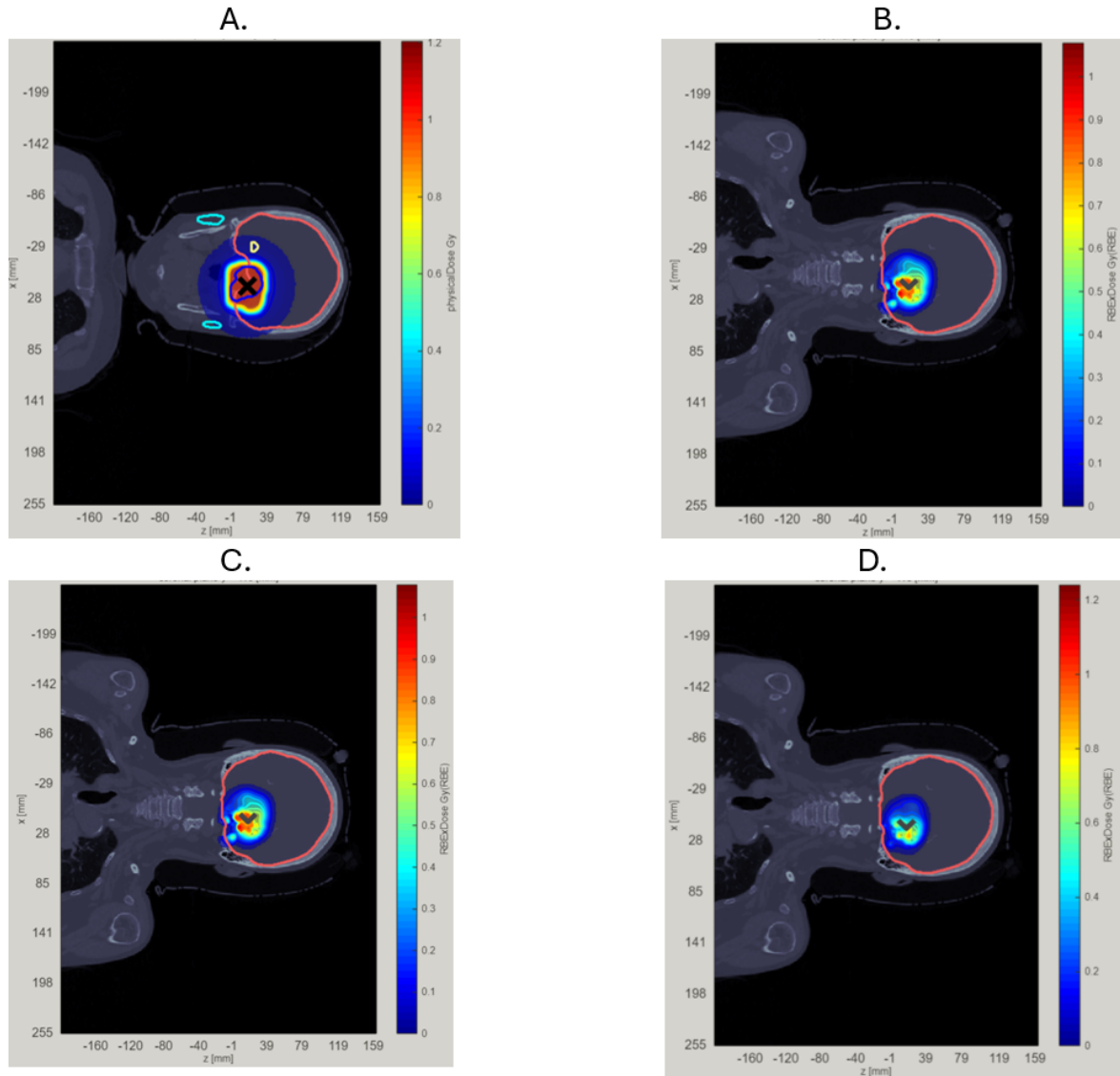
3.2 Treatment Planning Results: matRad Multi-Modality Study

Four plans were produced for each patient, one each for photons, protons, helium ions, and carbon ions, all with a two-beam configuration so that the comparison isolates the effect of the modality and not the beam arrangement. A conventional clinical five-beam photon plan was also available but is excluded here, since its different beam geometry would confound the modality contrast. Dose maps and dose–volume histograms are shown for the representative patient; the dose tables are reported for all three.

3.2.1 Dose Distributions

The dose maps (Figure 3.2) show the expected pattern: the photon plan deposits a broader low-dose envelope around the target, the proton plan tightens this envelope, and the helium and carbon plans confine the high dose most sharply to the target region.

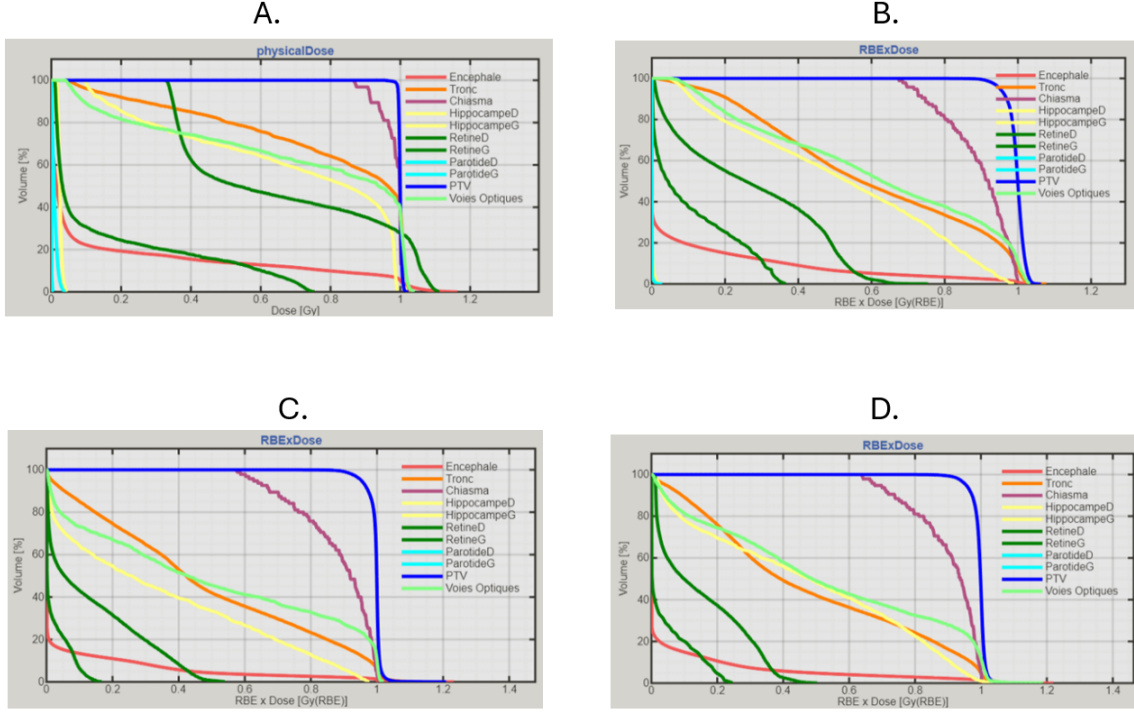
Figure 3.2: Dose distributions for the representative patient: the two-beam photon, proton, helium, and carbon plans (A–D). The photon plan shows the broadest spread of dose into the surrounding tissue, while the ion plans confine the dose more tightly around the target.



3.2.2 Dose–Volume Histograms

The dose–volume histograms (Figure 3.3) confirm that every plan covers the planning target volume. The plans separate on the organ-at-risk curves: the photon plan shows the largest organ-at-risk volumes at intermediate dose, the proton plan improves on this, and the helium plan shows the steepest fall-off and the greatest sparing.

Figure 3.3: Dose–volume histograms (DVH) for the representative patient: two-beam photon, proton, helium, and carbon plans (A–D). All plans achieve full planning target volume coverage; the plans differ mainly in the dose carried by the organs at risk at intermediate dose levels.



3.2.3 Mean and Maximum Doses

Table 3.4 and Table 3.5 list the mean and maximum doses to the planning target volume and the organs at risk for the four plans of each patient. The values are relative to the prescription, so the target mean is close to 1. The photon plan reports physical dose and all particle plans report RBE-weighted dose, so the modalities share a common effective-dose scale (Section 2.2).

Table 3.4: Mean dose to the planning target volume (PTV) and organs at risk, relative to prescription, for the two-beam photon, proton, helium, and carbon plans of the three patients. Photon values are physical dose; all particle values are dose weighted by the relative biological effectiveness (RBE).

Structure	Patient 1				Patient 2				Patient 3			
	Phot.	Prot.	Hel.	Carb.	Phot.	Prot.	Hel.	Carb.	Phot.	Prot.	Hel.	Carb.
PTV	0.9997	0.9964	0.9943	0.9951	0.9991	0.9933	0.9907	0.9912	0.9999	0.9966	0.9784	0.9959
Chiasma	0.9805	0.8914	0.8788	0.9000	0.9996	1.0001	0.9976	0.9975	0.8606	0.7405	0.4916	0.6372
Voies Optiques	0.7104	0.6083	0.4891	0.5310	0.8904	0.7295	0.6498	0.6773	0.4494	0.3247	0.2003	0.2676
Hippocampe G	0.6837	0.5168	0.3436	0.4707	0.7407	0.2977	0.1361	0.1629	0.3022	0.1108	0.0341	0.0585
Hippocampe D	0.0248	0.0010	0.0001	0.0000	0.1870	0.1709	0.0490	0.0673	0.3619	0.2826	0.1339	0.1627
Tronc	0.7875	0.5963	0.4780	0.4785	0.9273	0.6409	0.5519	0.5336	0.2765	0.2588	0.1650	0.2121
Encéphale	0.1591	0.0917	0.0640	0.0677	0.2358	0.1171	0.0842	0.0822	0.3081	0.1657	0.1066	0.1204
Rétine G	0.6624	0.2684	0.1347	0.1503	0.5065	0.1298	0.0552	0.0791	0.1036	0.0003	0.0004	0.0029
Rétine D	0.1558	0.1047	0.0237	0.0428	0.4441	0.2263	0.1268	0.1146	0.0656	0.0290	0.0061	0.0083
Parotide G	0.0094	0.0002	0.0000	0.0000	0.0047	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Parotide D	0.0003	0.0000	0.0000	0.0000	0.0014	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 3.5: Maximum dose to the planning target volume (PTV) and organs at risk, relative to prescription, for the two-beam photon, proton, helium, and carbon plans of the three patients. Photon values are physical dose; all particle values are dose weighted by the relative biological effectiveness (RBE).

Structure	Patient 1				Patient 2				Patient 3			
	Phot.	Prot.	Hel.	Carb.	Phot.	Prot.	Hel.	Carb.	Phot.	Prot.	Hel.	Carb.
PTV	1.0243	1.0633	1.2096	1.1584	1.0215	1.0609	1.2837	1.2030	1.0313	1.0466	1.0429	1.1360
Chiasma	1.0031	0.9993	1.0157	1.0074	1.0039	1.0170	1.0088	1.0112	0.9448	0.9040	0.6959	0.8944
Voies Optiques	1.0421	1.0338	1.0207	1.1897	1.0190	1.0388	1.0339	1.0504	1.0323	1.0211	1.0010	1.0100
Hippocampe G	0.9927	0.9903	0.9783	1.0639	1.0042	0.8813	0.9000	0.8005	1.0449	0.7756	0.5037	0.5980
Hippocampe D	0.0431	0.0241	0.0008	0.0003	0.8631	0.7216	0.5132	0.6423	1.0337	0.9614	0.8021	0.9909
Tronc	1.0339	1.0783	1.1960	1.1372	1.0540	1.0410	1.4356	1.3427	1.0319	1.0405	1.0245	1.0331
Encéphale	1.1638	1.0783	1.2330	1.2204	1.2692	1.0609	1.4356	1.3427	1.2114	1.0466	1.0574	1.1965
Rétine G	1.1129	0.7553	0.5412	0.5036	1.0373	0.4181	0.2930	0.2877	0.3449	0.0058	0.0017	0.0229
Rétine D	0.7543	0.3669	0.1676	0.2484	0.8214	0.5126	0.5094	0.3452	0.6645	0.1896	0.0916	0.0932
Parotide G	0.0416	0.0295	0.0007	0.0009	0.0233	0.0001	0.0001	0.0000	0.0000	0.0001	0.0001	0.0000
Parotide D	0.0067	0.0000	0.0000	0.0000	0.0118	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Across all three patients the target is covered in every plan, and the organ-at-risk doses separate by modality much as the distributions and dose–volume histograms suggest. The photon plan delivers the highest mean doses to most organs, the proton plan lowers them, and the helium plan achieves the lowest mean doses to the majority of the organs at risk, with the carbon plan giving sharp target dose but slightly less organ sparing than helium.

3.3 Agreement Between Geometric Volume Coefficient and Delivered Dose

The geometric measure was then compared against the doses of Section 3.2, by the Spearman rank correlation between the organ-level Geometric Volume Coefficient and the mean delivered dose, computed separately for each modality and each patient over all ten organs at risk. The alternative hypothesis is directional—an organ with a higher coefficient is expected to receive a higher, not a lower, mean dose—so the correlation was tested one-sided throughout. The structures with a coefficient of zero represent the correct identification of organs that lie in the separation zone and are, in every plan, spared. They are retained rather than excluded, since removing the cases the measure predicts most confidently would bias the analysis, and their tied values were assigned averaged ranks.

Table 3.6: Spearman rank correlation coefficient between the organ-level Geometric Volume Coefficient (GVC) and the mean delivered dose, computed over all ten organs at risk, for each modality and each patient. In every case the correlation is positive, meaning that organs with a higher Geometric Volume Coefficient receive a higher mean dose, and all correlations are significant (one-tailed; ** $p < 0.01$).

Patient	Photon	Proton	Helium	Carbon
Patient 1	0.847**	0.821**	0.862**	0.870**
Patient 2	0.769**	0.853**	0.828**	0.828**
Patient 3	0.922**	0.960**	0.935**	0.935**

The correlation is strong, positive, and stable across the study (Table 3.6). The more entangled the organ, the more dose it receives. The coefficient ranges from 0.769 to 0.960 across the twelve patient–modality combinations, significant in every case ($p < 0.01$). Ordering the organs by their coefficient reproduces the ordering of their mean doses, and this holds for photons, protons, helium, and carbon, and for all three patients. The agreement between a contour-only descriptor and the delivered dose is the central result of the thesis. It shows that the coefficient captures the dose burden imposed by the anatomy, whatever the delivery technique, and gives a sound basis for the quality-assurance use proposed in Section 2.4.

3.3.1 Sensitivity to the Zero-Complexity Organs

Five of the ten organs in Patient 1, and four in Patients 2 and 3, carry a coefficient of exactly zero and receive a common averaged rank. The same organs also receive the lowest doses in every plan, so the tied block contributes concordant pairs and raises the correlation. The analysis was therefore repeated over the non-zero organs alone, where the descriptor is asked to rank the entangled organs among themselves and not only to separate the exposed structures from the unexposed ones.

Table 3.7: Sensitivity analysis: Spearman rank correlation coefficient between the organ-level Geometric Volume Coefficient (GVC) and the mean delivered dose, computed over the non-zero-GVC organs at risk only ($n = 5$ for Patient 1, $n = 6$ for Patients 2 and 3). Significance (one-tailed) is marked * $p < 0.05$ and ** $p < 0.01$; the remaining values are not significant at $p < 0.05$.

Patient	n	Photon	Proton	Helium	Carbon
Patient 1	5	0.600	0.700	0.700	0.700
Patient 2	6	0.771	0.943**	0.886**	0.886**
Patient 3	6	0.771	0.943**	0.829*	0.829*

The correlation remains positive throughout, between 0.600 and 0.943, so the direction of the relationship does not depend on the zero-complexity organs. Its magnitude and its significance do. The values fall for Patient 1 and Patient 3, and six of the twelve combinations reach $p < 0.05$, none of them in Patient 1. This is a question of statistical power and not of the relationship itself: with $n = 5$, a coefficient of at least 0.9, that is at most one adjacent rank inversion, is needed to reach $p < 0.05$, so a value of 0.700 cannot be significant however consistent the ordering appears. Table 3.6 should therefore be read as follows. Its correlations include the correct identification of the separated organs, which is itself a property the descriptor should have, while the finer ranking among the entangled organs is supported in direction but not yet established at conventional significance. Testing it properly requires more non-zero organs per case than three patients can provide, and this is one of the reasons for scaling to the full cohort (Section 5.1).

Chapter 4

Discussion and Conclusion

This chapter reads the results of Chapter 3 against the objective set out at the start: to lay the foundation for a quality-assurance process that can explain *why* a plan fails. It takes the descriptor first, then the planning study that supplies the independent check, then what the formulation enables.

4.1 Tumour Geometric Complexity as a Geometric Descriptor: Interpretation and Limitations

The central result is the agreement of Section 3.3. The contour-only ordering of the organs reproduces the ordering of their delivered mean doses, for every modality and every patient tested. The coefficient is computed before any plan exists and independently of the beam, so the agreement is not circular: a purely geometric quantity recovers the dose burden imposed by the anatomy, whatever the delivery technique. This supports the descriptor’s central claim, that it measures intrinsic difficulty and not the outcome of one particular optimisation.

The patient-level aggregation (Eq. 2.8) behaves as intended: the most entangled structures dominate the score while the low-complexity organs still contribute a small, non-zero amount instead of being averaged away. The three patients also span a wide range of difficulty, from a well-separated case to one in which the optic chiasm is almost fully enclosed by the target. The measure discriminates between geometries and does not saturate.

Three limitations qualify these findings. First, the results rest on three patients, and part of the strength of the correlation comes from the organs lying entirely in the separation zone, which take a coefficient of exactly zero and the lowest dose and so enter the ranking as concordant tied pairs. Restricted to the non-zero organs, the correlation stays positive throughout (0.600–0.943) but falls, and reaches significance in only half of the twelve combinations, since five or six organs then remain (Section 3.3.1). The agreement holds as a proof of concept. Its strength, and in particular the descriptor’s ability to rank the entangled organs among themselves, can only be established on the full cohort (Section 5.1).

Second, the descriptor depends on modelling choices that were fixed here and not validated: the penalties of Eq. 2.3, the threshold distance of Eq. 2.2, derived from a single achievable dose gradient, and the clinical priority weights, based on published tolerance data instead of derived

from a formula. The mathematical form of the coefficient is based on physical reasoning and clinical intuition, as stated in Section 2.1, and other forms satisfying the same ordering are possible.

Third, a limitation intrinsic to the descriptor’s construction. It captures proximity to the target but not beam orientation. Two organs at the same distance can receive very different doses when one lies along a beam direction and the other does not, so structures that extend along the beam axis, such as the brainstem, or that sit in the entrance path, such as the retina, can receive dose out of proportion to their geometric coefficient. This is visible in the present results, and it accounts for part of the departure of the correlation from unity. The effect is known: Yuan et al. [22] addressed it by modifying the Euclidean distance to reflect beam geometry. A beam-aware distance is therefore a natural extension, noted in Section 5.1.

4.2 Treatment Planning Across Modalities: What We Learned

For this thesis the planning study is a means and not an end. Read as a dosimetric comparison it reproduces the pattern expected from beam physics: every plan covered the target, while the organ-at-risk doses fell from photons through protons to the helium and carbon plans. Its value here is that these organ doses depend strongly on where each organ lies relative to the target, the same anatomical difficulty the descriptor captures without a plan, and this is what makes it usable as the independent check of Section 3.3.

One qualification applies to reading the comparison dosimetrically. The plans are research plans, simplified as described in Section 2.2.3, and the three ion plans use different biological models, a constant RBE for protons, the Local Effect Model for carbon, and a LET-based parameterisation for helium, which are not calibrated against one another. A statement that one ion spares an organ better than another therefore carries the uncertainty of those models and of the simplified planning, and the ion-versus-ion ordering should be read with that in mind. The analysis of Section 3.3 is unaffected, since it compares organs within a plan and not plans with each other.

One case is particularly instructive. In Patient 2 the optic chiasm was almost entirely overlapped by the target, and no modality, not even the ion plans, could spare it: its mean dose stayed close to prescription across all four. When the anatomy leaves an organ geometrically inseparable from the target, the choice of modality cannot recover the goal. This is the situation the descriptor is meant to flag in advance, and it did: the chiasm carried the highest Geometric Volume Coefficient in the study, and the dose confirmed that the coefficient reflected an unavoidable, anatomy-imposed burden and not a shortcoming of the plan.

4.3 Towards a Richer Quality-Assurance Framework: What the Formulation Enables

A validated geometric descriptor supplies the piece the binary verdict lacks, and Patient 2’s chiasm is a concrete instance of the anatomy-limited failure it identifies. By grading case difficulty on a reproducible scale, Tumour Geometric Complexity turns the distinction between a sub-

optimal and a complexity-limited plan into a decision that no longer rests on expert judgement alone, as the four-class scheme of Section 2.4 sets out.

Three things follow. The outcome becomes explainable, since the organ-level coefficient identifies which structures drive the difficulty while the patient-level score grades the case. The process becomes more reproducible and easier to scale, since the grade is computed from contours by a fixed procedure, which should reduce the variation between centres. And the coefficient is a natural feature for automation: the same number that grades a case by hand can enter a learned classifier that predicts the four classes directly. The scheme’s credibility rests on the geometry–dose agreement; its validation against clinical decisions is future work (Section 5.2).

4.4 Overall Conclusions

This thesis set out to make radiotherapy quality assurance able to explain its verdicts. Current review is manual, expert-dependent, and returns a verdict that cannot distinguish a poorly made plan from an anatomy that never permitted a good one, because case difficulty is nowhere quantified. The work presented here is a proof of concept for quantifying it.

It contributes a geometric descriptor of case difficulty, the Tumour Geometric Complexity, computed from the delineated contours at the organ and patient level; a multi-modality matRad planning study that supplies the delivered doses; and the demonstration, across three patients, that the contour-only descriptor tracks those doses across all modalities tested. On that basis it proposes a four-class scheme that separates plans limited by poor planning from plans limited by anatomy.

These findings are a proof of concept established on three cases. The coefficient is a descriptor grounded in physical reasoning rather than a derived quantity, its parameters were fixed rather than fitted, and the four-class scheme has not yet been tested against QA decisions. What the work establishes is that case difficulty can be graded from the contours alone, reproducibly, and that the grade tracks the dose the anatomy imposes. That is the foundation on which cohort-scale validation and the learning-based automation of Chapter 5 can be built.

Chapter 5

Future Work

The work presented here is a proof of concept. Three steps would turn it towards a usable tool: scaling the computation to a full cohort, validating the four-class scheme against clinical practice, and automating the classification. Together they form the geometric foundation for the doctoral continuation this thesis initiates.

5.1 Scaling to the Full Patient Cohort

The immediate next step is to compute the Geometric Volume Coefficient across a larger multi-institutional cohort, so that the organ- and patient-level values are characterised as distributions and not as the handful reported here. This would show how the descriptor behaves over the natural range of anatomies met in practice, and would give the agreement of Section 3.3 the number of non-zero organs that the ranking among entangled structures requires. It would also allow the model parameters to be tested for robustness at scale. The penalties of Eq. (2.3), fixed here at the simplest instance satisfying the required ordering, could then be optimised against dose rather than assumed. The threshold distance d_c of Eq. (2.2) was derived from a single dose gradient; its stability across patients, prescription levels, and delivery systems must be quantified, and the gradient may need to be defined per modality instead of once. The clinical priority weights w_{priority} should be examined for sensitivity against the institutional (Baclesse) protocols and refined as larger datasets become available, and the patient-level aggregation revisited so that the score stays comparable across cases with differing numbers of organs at risk.

5.2 Validating the Four-Class Framework

The four-class scheme of Section 2.4 is supported here by the agreement between geometry and dose, but it has not been tested against QA decisions on real plans. Its validation requires a cohort of trial plans with recorded QA verdicts, which today means conventionally delivered photon treatments; the agreement of Section 3.3 across modalities is what allows a scheme calibrated on such a cohort to carry over to particle therapy. The central question is whether the cases the framework labels complexity-limited coincide with those that expert reviewers accept on clinical grounds despite a nominal deviation. Establishing that correspondence would

show that the descriptor recovers, automatically, a judgement currently made case by case. This stage must also fix the operating point separating low from high complexity at the branch of the decision flow.

Once validated, the classification can be automated, since the organ- and patient-level coefficient is a natural geometric feature for a learned model that predicts the four classes from routinely available data. This is the technical core of the doctoral continuation, which would extend the geometric foundation laid here towards an end-to-end, automated QA pipeline for multi-institutional trials. A longer-term aim is to move QA from a compliance check towards a prognostic one, relating the complexity grade to clinical outcomes such as tumour control, survival, and toxicity. The descriptor then becomes one input to a QA process that is faster and more reproducible than expert review, and able to explain its verdicts, which is the aim with which this thesis began.

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Appendix A

Skills and Knowledge Acquired

I came to the project with the dosimetry and radiotherapy physics of my Master's coursework, and with no practical experience of any of it. Treatment planning is the clearest instance. I knew what a dose distribution was; I had never produced one. Working through matRad case by case taught me what inverse planning is in practice: the planner places objectives on the target and the organs, and an optimiser searches for the fluence that best satisfies them. The related skill was data handling. I used the Python library pydicom to read the planning CT and the RTSTRUCT contours, and to convert them into the masks the geometric computation operates on. The computational side was the part I found most familiar, and it was not the difficulty of the project.

The physics of the modalities I had met in lectures, but not with the specificity a plan demands. Knowing what linear energy transfer is differs from knowing why protons take a constant relative biological effectiveness, carbon ions the Local Effect Model, and helium ions a parameterisation fitted to data. The same holds for physical dose, biological effect, RBE-weighted dose and biologically effective dose. I could define each. I could not have said which to optimise on, or why the identical fractionation across my four plans is what makes that choice defensible. That judgement I acquired here.

The larger change was in how I understand the problem the physics is meant to solve. I learnt what quality assurance is in a clinical trial: a plan is reviewed against the protocol before delivery, by experts, and that review is what keeps a multicentre trial comparable across institutions. I also learnt that planning is a negotiation and not a search for a correct answer. Covering the target and sparing an organ compete directly, and a good plan is a defensible compromise. Sometimes the compromise is not available, because the anatomy has settled it in advance. That is where the question of this thesis came from.